温州肯恩大学
WENZHOU-KEAN UNIVERSITY

Course Information

Course Title: Foundations of Business Analytics
Course Number and Section: MGS 3101 W03
Campus Location: Wenzhou-Kean University
Semester: 2025 Fall
Meeting Days and Times: Tue & Thu 5:30pm - 6:45pm
Meeting Location: CBPM A202

Instructor Information

Instructor Name: Dawei CHEN
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Office Hours: Mon & Wed 4pm – 6pm | Fri 4pm –5pm
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Course Description

Foundations of Business Analytics in an introductory course that demonstrates and elaborates the power of continuous iterative analysis of past business performance on decision making across different domains using python. The course equips the students with the skills and techniques that they need for the Business Analytics Program. Through the course; students explore the applications of Business Analytics; Business Intelligence; Data Mining; and Big Data from a managerial perspective.

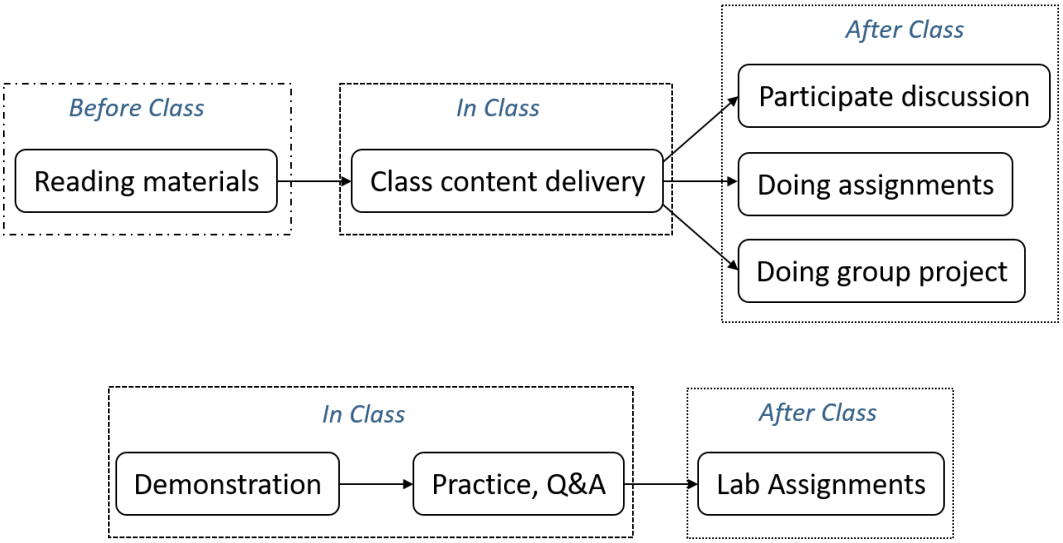
Intended Learning Outcomes

Upon successful completion of this course, the student should be able to:

- Demonstrate the motivation, drivers, and importance of business analytics.
- Describe the different types of analytics and illustrate the objectives of each one of them.
- Discuss the importance of data visualization and apply the different types of visualization techniques using Python.
- Identify some of the emerging technologies that may impact analytics, business intelligence, and data science such as Internet of Things (IOT), cloud computing.
- Outline the ethical and legal issues related to deployment of analytics.

- Identify main characteristics, skills, and requirements of analytics-related career opportunities.
- Communicate the results of a business analytics project to a non-technical audience of decision-maker and stakeholders.
- Able to independently complete beginner-level Kaggle competitions.

Instructional Method



Course Materials

Textbook:

- Business Analytics, 5th Edition, by Camm et al., Cengage 2023.
- Python for Data Analytics, 3rd Edition, by Wes Mckinney, O'Reilly 2022.

(*Additional reading materials may be introduced throughout the semester.*)

Schedule (tentative)

Date		Topics/Readings/Resources	Assignments
W1	09/02	Course Introduction & House Rule	
	09/04	Introduction to Business Analytics (Camm, Ch1)	
W2	09/09	Python Basics & Jupyter Notebook I (Wes, Ch2)	
	09/11	Python Basic & Jupyter Notebook II (Wes, Ch3)	
W3	09/16	Pandas I (Wes, Ch5, S5.1)	
	09/18	Pandas II (Wes, Ch5, S5.2)	
W4	09/23	Descriptive Statistics (Camm, Ch2)	Assignment 1: 09/26 (Introductions to BA)
	09/25	(lab) (Wes, Ch5, S5.3)	
W5	09/30	Data Visualization (Camm, Ch3)	
	10/02	(no class)	
W6	10/07	(no class)	
	10/09	(lab) (Wes, Ch9)	
W7	10/14	Data Wrangling I (Camm, Ch4)	Assignment 2: 10/18 (Desc. Stat. + Data Visu.)
	10/16	Review for Midterm Exam	
W8	10/21	Midterm Exam (computer-based)	
	10/23	Data Wrangling II (lab) (Wes, Ch7 & Ch8)	
W9	10/28	Statistical Inference I: Sampling and Estimation	
	10/30	Statistical Inference II: Interval Estimation	
W10	11/04	Statistical Inference III: Hypothesis Testing	
	11/06		
W11	11/11	Simple Linear Regression (Camm, Ch8)	Assignment 3: 11/14 (Data Wran. + Stat Infe.)
	11/13		
W12	11/18	Multiple Linear Regression I (Camm, Ch8)	
	11/20		
W13	11/25	Multiple Linear Regression II (Camm, Ch8)	
	11/27		
W14	12/02	Time Series Analysis (Camm, Ch9)	
	12/04		
W15	12/09	Q&A for Group Project	
	12/11	Group Project Presentation	
W16	12/16	Review, Q&A for Final Exam	
	12/18	Final Exam (written)	

Assessment

Grade Breakdown	
Participation	10%
Assignment	3x5%=15%
Midterm Exam	25%
Final Exam	25%
Group Project	25%
Total	100%

- **Participation:** Class participant is mandatory according to school policy. **1 point** will be deducted for each class absence without approval.

- **Assignment:** There will be 3 assignments throughout the semester, each worth 5% of the final grade. A 50% penalty will be applied for late submission.

- **Midterm Exam**: A computer-based test will be conducted jointly with another class (MGS 3101 W02) to ensure consistency and fairness in grading across both classes. The exam date and time will be rescheduled to avoid conflict with students' other courses. No make-up exam will be offered!

- **Final Exam**: A written test will be held jointly with another class (MGS 3101 w02) to ensure consistency and fairness in grading across both classes. The exam date and time will be rescheduled to avoid conflict with students' other courses. No make-up exam will be offered!

- **Group Project**: Groups of 4 students will be randomly formed to work on a real-world business analytics project. Information regarding group assignments, specific project guidelines, and datasets will be provided on October 30, 2025 (tentative). Several project options will be available, and each group can select the one that best aligns with their interests. At the end of the semester, each group will have 20 mins to present their project + 5 mins Q&A. Detailed assessment criteria for the group project will be shared at a later time.